



Team 2 - Asset monitoring and change detection

Objective

The objective of this project is to continuously monitor an organization's digital assets, detect any new assets or changes in existing ones, and identify potential security risks in real time. It also provides alerts and a dashboard to help security teams quickly understand and respond to changes in the attack surface.

Key Responsibilities

1. Asset Discovery & Inventory Management

- Identify and maintain a list of all external assets.
- Track domains, subdomains, IPs, services, and cloud resources.
- Keep asset inventory updated automatically.

Example:

example.com
api.example.com
mail.example.com
192.168.1.10

2. Continuous Asset Monitoring

- Perform scheduled scans automatically.
- Monitor assets for changes without manual intervention.
- Ensure continuous visibility of the attack surface.

Example:

Daily Nmap Scan
Hourly DNS Check
Weekly Vulnerability Scan

3. Change Detection

- Compare previous and current asset states.
- Identify newly exposed services, ports, subdomains, or configuration changes.
- Detect unauthorized modifications.



Example:

Previous: Port 80 Open

Current: Ports 80,443 Open

Detected Change:

New Port 443 Added

4. Vulnerability & Exposure Monitoring

- Detect vulnerabilities and misconfigurations.
- Identify exposed services and risky configurations.
- Monitor attack surface growth.

Example:

Critical CVE Found

Weak SSL Configuration

Exposed Admin Panel

5. Risk Assessment

- Classify detected changes based on severity.
- Prioritize security issues.
- Reduce false positives.

Example:

Change	Severity
New Port	Medium
New Vulnerability	High
DNS Hijack Risk	Critical

6. Alert Generation

- Notify security teams when significant changes occur.
- Support multiple notification channels.
- Enable rapid incident response.

Example:

ALERT:

New Vulnerability Detected

Asset: api.example.com

Severity: High



7. Historical Data Management

- Store scan results and snapshots.
- Maintain asset history.
- Support trend and forensic analysis.

Example:

June 1 → Port 80

June 5 → Port 80,443

June 10 → Vulnerability Detected

8. Visualization & Reporting

- Display assets, vulnerabilities, and alerts through dashboards.
- Provide real-time monitoring views.
- Generate security reports.

Example Dashboard Widgets

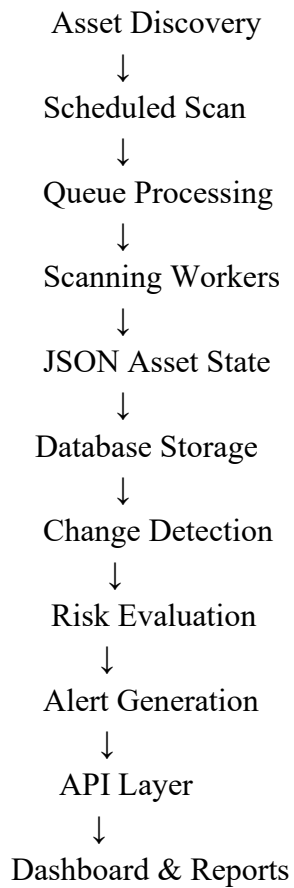
- Total Assets
- New Changes
- Open Vulnerabilities
- Critical Alerts
- Scan Status

Architecture Layers

Layer	Responsibility
Scheduler	Trigger periodic scans
Queue	Manage scan jobs asynchronously
Workers	Execute security scanning tools
Asset State Builder	Standardize scan output into JSON
Storage	Store asset snapshots and relationships
Change Detection	Compare current vs previous states
Risk Analysis	Prioritize detected changes
Alert Engine	Generate notifications
API Layer	Expose services and data
Dashboard	Visualize security posture
Observability	Monitor platform health



Data Flow



Asset Monitoring API Standard Structure

1. Asset Scan Request API

Input JSON

```
{
  "asset_id": "AST-1001",
  "domain": "example.com",
  "scan_type": "full_scan",
  "tools": [
    "nmap",
    "httpx",
    "nuclei"
  ],
  "schedule": {
```



```
"enabled": true,  
  "interval": "6h"  
},  
"priority": "high",  
"requested_by": "security_team"  
}
```

Output JSON

```
{  
  "status": "success",  
  "message": "Scan task added to queue",  
  "task_id": "TASK-98765",  
  "queued_at": "2026-05-29T12:30:00Z",  
  "queue_engine": "Redis"  
}
```

2. Worker Scan Result Structure

Standardized Worker Output

```
{  
  "task_id": "TASK-98765",  
  "asset_id": "AST-1001",  
  "scan_results": {  
    "open_ports": [  
      {  
        "port": 80,  
        "service": "HTTP",  
        "status": "open"  
      },  
      {  
        "port": 443,  
        "service": "HTTPS",  
        "status": "open"  
      }  
    ],  
    "subdomains": [  
      "api.example.com",
```



```
    "mail.example.com"
  ],
  "vulnerabilities": [
    {
      "tool": "nuclei",
      "severity": "high",
      "cve": "CVE-2025-1234",
      "description": "Exposed admin panel"
    }
  ]
},
"scan_completed_at": "2026-05-29T12:45:00Z"
}
```

3. Asset State Builder JSON

Structured Asset Schema

```
{
  "asset_id": "AST-1001",
  "asset_type": "domain",
  "primary_domain": "example.com",
  "ip_addresses": [
    "192.168.10.20"
  ],
  "services": [
    {
      "port": 443,
      "protocol": "tcp",
      "service_name": "https"
    }
  ],
  "technologies": [
    "nginx",
    "react"
  ],
  "last_scan": "2026-05-29T12:45:00Z",
  "risk_score": 8.5
}
```



4. Change Detection Engine JSON

Previous vs Current Comparison

```
{
  "asset_id": "AST-1001",
  "change_detected": true,
  "changes": [
    {
      "type": "new_open_port",
      "old_value": null,
      "new_value": 8080,
      "severity": "medium"
    },
    {
      "type": "dns_change",
      "old_value": "192.168.10.20",
      "new_value": "192.168.10.30",
      "severity": "high"
    }
  ],
  "detected_at": "2026-05-29T13:00:00Z"
}
```

5. Alert Engine JSON

Alert Structure

```
{
  "alert_id": "ALT-2026-001",
  "asset_id": "AST-1001",
  "severity": "critical",
  "title": "New Critical Vulnerability Detected",
  "description": "CVE-2025-1234 detected on api.example.com",
  "notification_channels": [
    "email",
    "slack",
    "siem"
  ],
  "created_at": "2026-05-29T13:05:00Z",
  "status": "sent"
}
```



6. Dashboard API Response JSON

Frontend Dashboard Data

```
{
  "summary": {
    "total_assets": 120,
    "active_assets": 115,
    "critical_alerts": 4,
    "high_risk_assets": 12
  },
  "recent_changes": [
    {
      "asset": "api.example.com",
      "change": "New Open Port 8080",
      "severity": "medium"
    }
  ],
  "vulnerability_stats": {
    "critical": 4,
    "high": 10,
    "medium": 20,
    "low": 15
  }
}
```

7. API Standard Response Format

Common Response Schema

```
{
  "status": "success",
  "message": "Operation completed",
  "data": {},
  "errors": null,
  "timestamp": "2026-05-29T13:10:00Z"
}
```




Importance of OWASP Top 10 Integration in Asset Monitoring & Change Detection

Integrating the **OWASP Top 10** into an Asset Monitoring & Change Detection platform significantly improves the system's ability to identify, prioritize, and remediate security risks. Instead of only detecting asset changes, the platform can assess the **security impact** of those changes based on industry-recognized web application risks.

- Risk-Based Change Prioritization
- Enhanced Vulnerability Classification
- Improved Attack Surface Visibility
- Accurate Vulnerability Scoring
- Automated Remediation Recommendations
- Better Alerting and Reporting
- Compliance and Security Best Practices
- Support for CTEM and Attack Surface Management

Deliverables from Team 2

- **Asset Inventory Report**

Provides a continuously updated list of discovered assets.

Example,

```
{
  "asset_id": "AST001",
  "asset_type": "Subdomain",
  "asset_name": "api.company.com",
  "ip": "192.168.1.10",
  "status": "Active"
}
```

- **Asset Exposure Report**

Provides visibility into externally exposed services.

Example,



```
{
  "asset": "api.company.com",
  "open_ports": [80, 443, 8080],
  "services": ["HTTP", "HTTPS"],
  "exposure_level": "Public"
}
```

- **Change Detection Report**

Identifies newly discovered assets and configuration changes.

Example,

```
{
  "asset": "api.company.com",
  "change_type": "NEW_PORT",
  "old_state": [80],
  "new_state": [80, 443],
  "timestamp": "2026-06-02"
}
```

- **Vulnerability Discovery Feed**

Provides findings collected from Nuclei, OpenVAS, and other scanners.

Example,

```
{
  "asset": "api.company.com",
  "finding": "Exposed Admin Panel",
  "severity": "High",
  "source": "Nuclei"
}
```

- **Attack Surface Intelligence**

Provides a complete view of the organization's external attack surface.

- **Risk Context Data**

Provides contextual information for risk scoring.



Example,

```
{  
  "asset": "payment.company.com",  
  "business_criticality": "Critical",  
  "environment": "Production",  
  "internet_facing": true  
}
```

Conclusion

The Asset Monitoring & Change Detection system provides a comprehensive and automated approach to discovering, monitoring, and securing an organization's digital assets. By continuously tracking domains, subdomains, IP addresses, services, cloud resources, and configuration changes, the platform enables real-time visibility into the organization's attack surface.

The solution combines asset discovery, continuous monitoring, change detection, vulnerability and exposure monitoring, risk assessment, alert generation, historical data management, and dashboard-based reporting into a unified architecture. This ensures that security teams can quickly identify newly exposed assets, detect unauthorized changes, assess potential risks, and respond effectively to emerging threats.

The integration of OWASP Top 10 principles further enhances the platform by enabling risk-based vulnerability prioritization, accurate vulnerability scoring, automated remediation recommendations, and improved security reporting. This helps organizations focus on the most critical security issues and strengthen their overall security posture.

Through standardized APIs, scalable event-driven architecture, automated scanning workflows, and actionable intelligence delivered to Vulnerability & Risk Analysis teams, the system serves as a key component of modern Continuous Threat Exposure Management (CTEM) and Attack Surface Management programs. Ultimately, it enables proactive security operations, reduces attack surface risks, and supports informed decision-making for effective cybersecurity governance.